Lesson 17: Bootstrap Hypothesis Testing

BSTA 551: Statistical Inference

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Review: Where We Left Off

Lesson 16: Further Aspects of Hypothesis Testing; CI-Test Duality

- Statistical vs. clinical significance: always report effect sizes + CIs
- **Duality Theorem:** $\theta_0 \in$ CI $\iff$ fail to reject $H_0 : \theta = \theta_0$
- Inverting a test gives a general CI construction method
- Equivalence and non-inferiority testing via TOST / one-sided bounds

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Today's Goals:

1. Understand the conceptual framework for **bootstrap hypothesis testing** (C&H Ch. 8)
2. Implement the **one-sample bootstrap test** by resampling under H_0
3. Implement the **two-sample bootstrap test**
4. Understand **permutation tests** as an exact alternative to parametric tests
5. Compare parametric, bootstrap, and permutation tests: when does each apply?
6. Apply these methods to real medical data scenarios

Roadmap

Topic	Key Idea	Reference
Why bootstrap testing?	When parametric assumptions fail	C&H Ch. 8
One-sample bootstrap test	Resample after shifting to H_0	C&H 8.1
The naive bootstrap (wrong)	Resampling without shifting gives invalid null	C&H 8.1
Bootstrap p-value	Fraction of bootstrap statistics exceeding observed	C&H 8.1
Bootstrap vs. t-test	Robustness for non-normal, small samples	C&H 8.1

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Why This Matters

Parametric tests (z, t) require distributional assumptions that may not hold in practice. The bootstrap test provides a **distribution-free** alternative that is valid under much weaker conditions — essential for small samples, skewed outcomes, and novel test statistics in biomedical research.

The Bootstrap Hypothesis Test Framework (C&H Ch. 8)

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We have used the bootstrap to **construct confidence intervals** (Lessons 9–10). Now we use it for **hypothesis testing**. The ideas are related but not identical.

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Resample from estimated distribution	Resample from distribution **under** H_0
Goal: capture sampling variability of $\hat{\theta}$	Goal: approximate null distribution of test statistic
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Core Principle (C&H 8.1)

A bootstrap hypothesis test approximates the **null distribution** of the test statistic by resampling from a population that **satisfies** H_0. The p-value is the fraction of bootstrap statistics at least as extreme as the observed value.

The Bootstrap Test: General Algorithm

Setup: Observe $x_1, \ldots, x_n$. Test $H_0 : \theta = \theta_0$ vs. $H_a : \theta \neq \theta_0$. Let $T = T(x_1, \ldots, x_n)$ be a test statistic.

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Algorithm:

1. Compute the observed test statistic $T_{\text{obs}} = T(x_1, \ldots, x_n)$
2. **Shift the data** to satisfy H_0: define $x_i^* = x_i - \bar{x} + \theta_0$ (so the shifted sample has mean exactly θ_0)
3. For $b = 1, \ldots, B$:
   - Draw a bootstrap sample $x_1^{*b}, \ldots, x_n^{*b}$ **with replacement** from $(x_1^*, \ldots, x_n^*)$
   - Compute $T^{*b} = T(x_1^{*b}, \ldots, x_n^{*b})$
4. Compute p-value:

$$\hat{p} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}(|T^{*b}| \geq |T_{\text{obs}}|)$$

Why Shift the Data?

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Shifting: By replacing x_i with $x_i^* = x_i - \bar{x} + \theta_0$, we create a dataset that:

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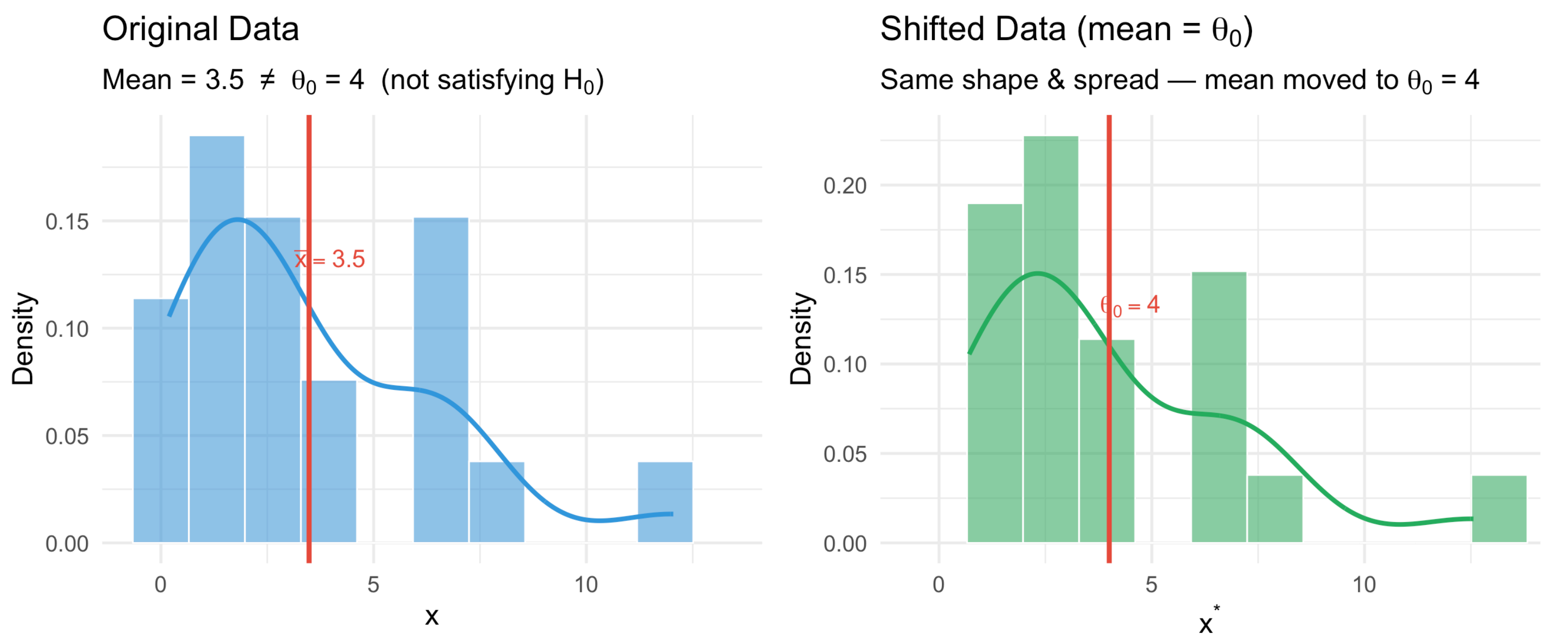
▶ Code



One-Sample Bootstrap Test (C&H 8.1)

Full Implementation: One-Sample Bootstrap Test

Medical Context: A hospital monitors C-reactive protein (CRP, mg/L) as an inflammation marker. The healthy adult population norm is mean CRP = 1.5 mg/L. We observe a small sample of post-surgical patients and test whether their mean CRP differs from this norm.

$$H_0 : \mu = 1.5 \quad \text{vs} \quad H_a : \mu \neq 1.5$$

▶ Code

```
Sample mean CRP: 2.083 mg/L
```

▶ Code

```
Sample SD:       1.588 mg/L
```

▶ Code

```
Observed t-statistic: 1.7238
```

▶ Code

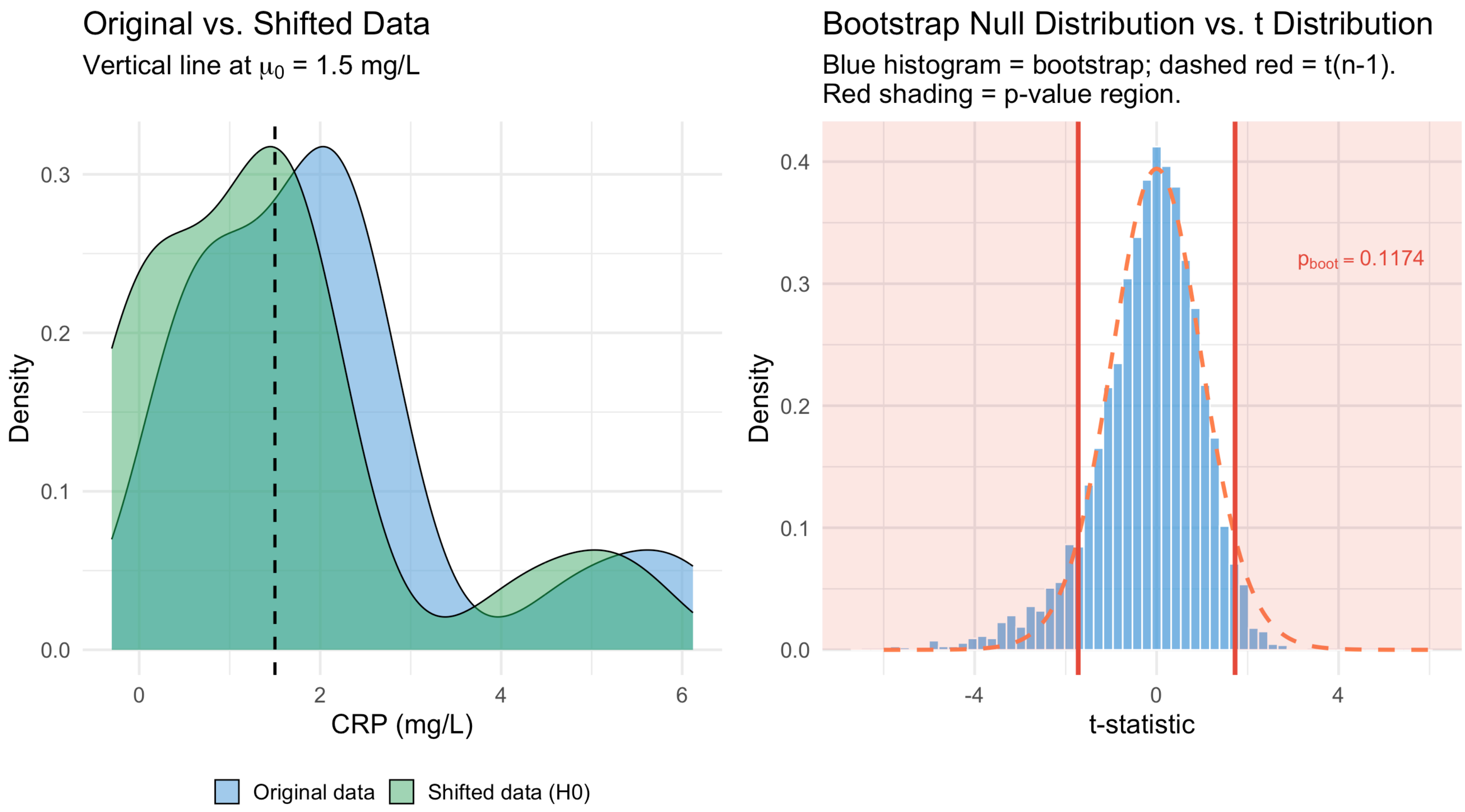
```
Bootstrap p-value:    0.1174
```

▶ Code

```
Parametric t-test p:  0.0994
```

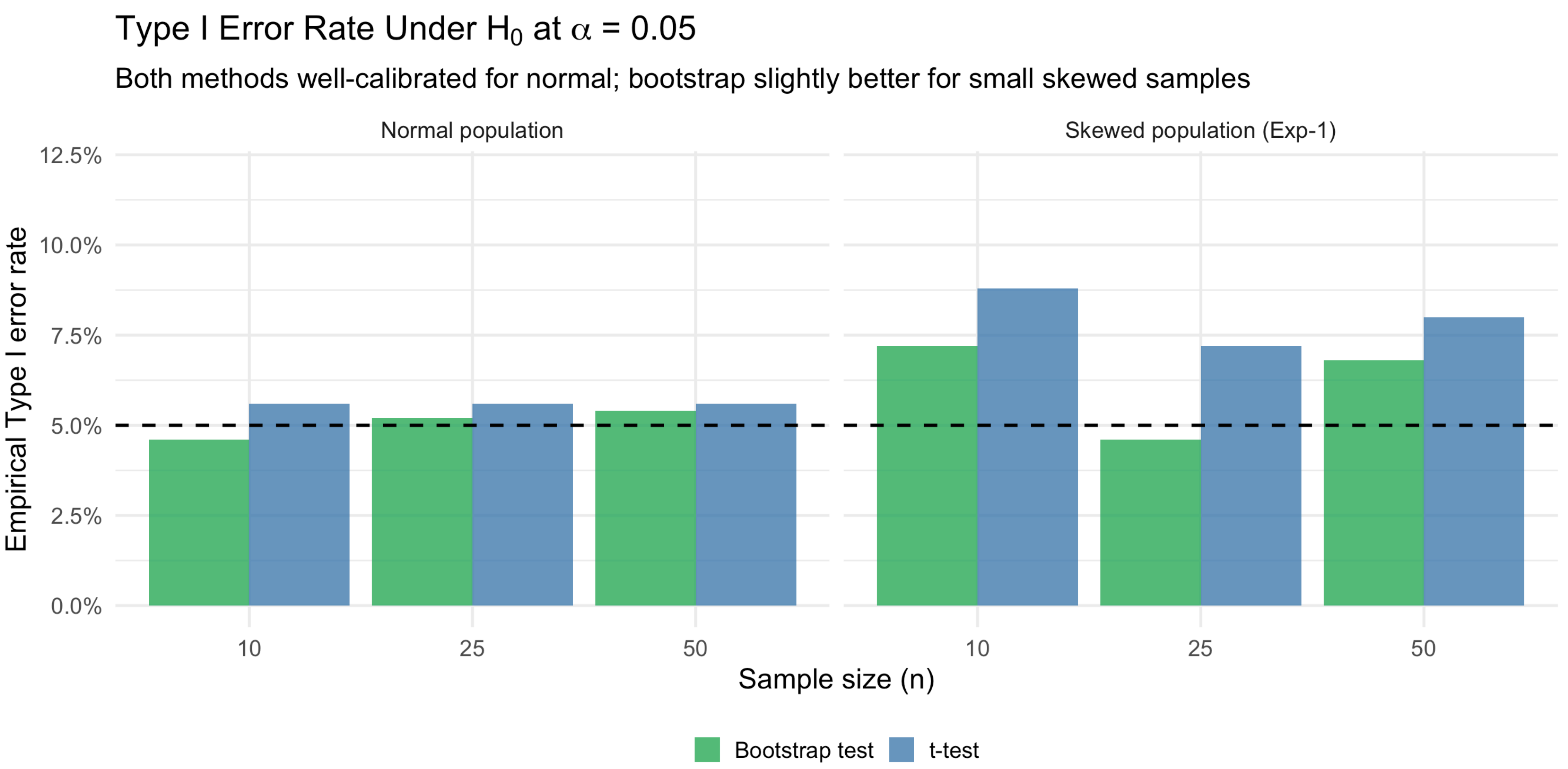
Visualizing the Bootstrap Null Distribution

▶ Code



When Does the Bootstrap Test Outperform the t-Test?

▶ Code



What Happens If You Don't Shift? (A Common Mistake)

It is tempting to simply bootstrap from the **original** data and compare to θ_0. Why is this wrong?

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⚠ The Naive Bootstrap Test — Incorrect

Resample from the original $x_1, \ldots, x_n$ and compute $\bar{x}^{*b}$.
Compare the fraction of $\bar{x}^{*b}$ farther from $\bar{x}$ than $|\bar{x} - \theta_0|$.

Problem: The bootstrap distribution is centered at $\bar{x}$, not at θ_0.
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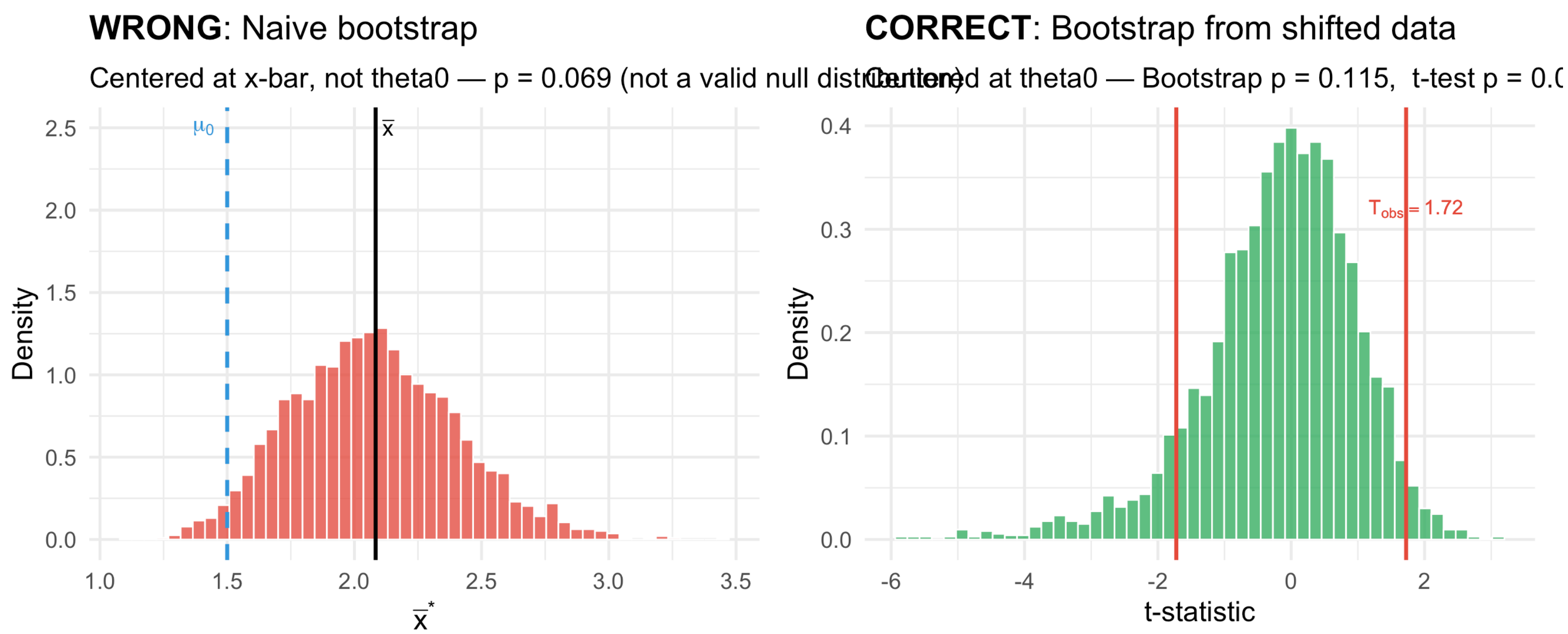
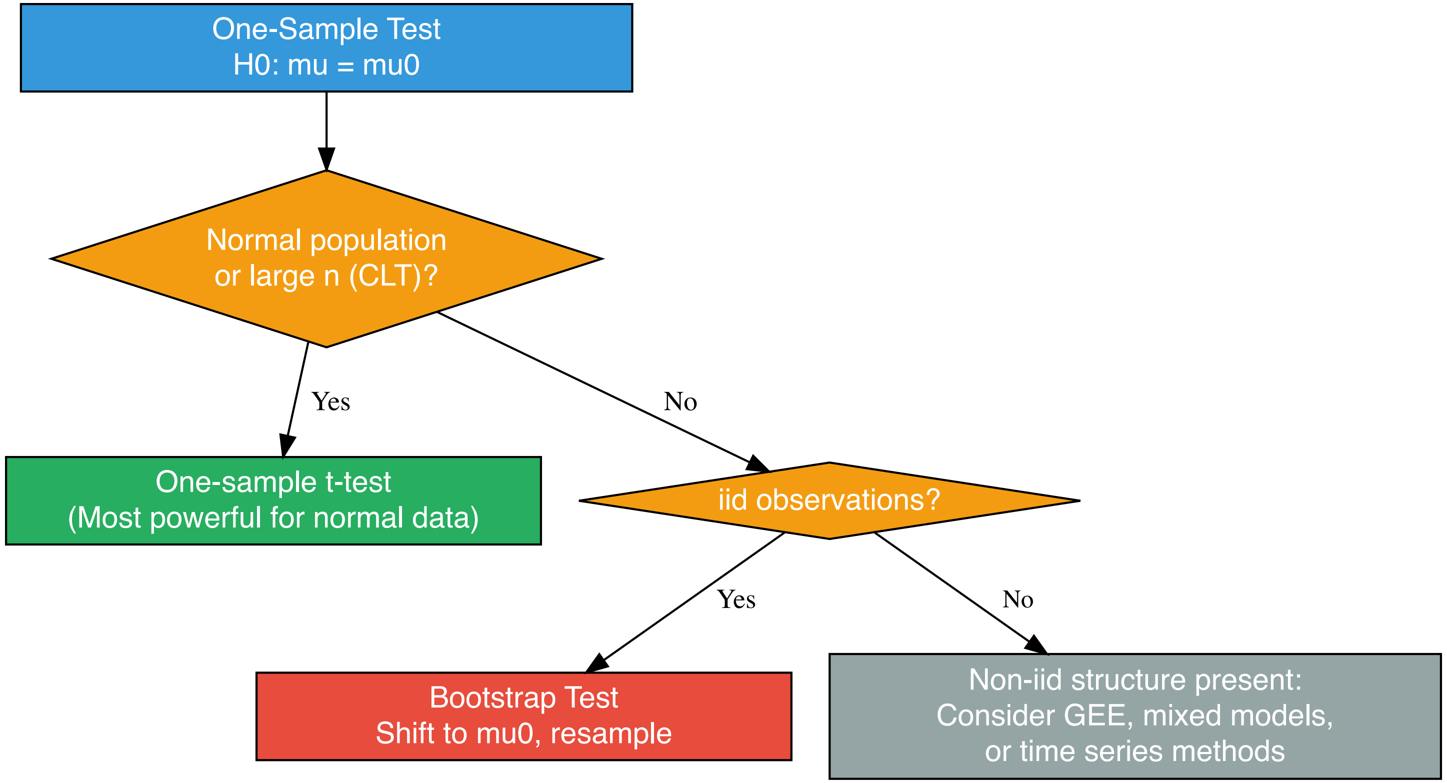


Choosing Between Methods: One-Sample Tests

Decision Framework

For a **one-sample** test $H_0 : \mu = \mu_0$, which method should you use?

▶ Code



Assumptions Compared

Assumption	t-test	Bootstrap test
Normal population (or large n)	Required	Not required
iid observations	Required	Required
Distribution-free validity	No	Approximately
Works for any test statistic	No	**Yes**
Computationally intensive	No	Moderate

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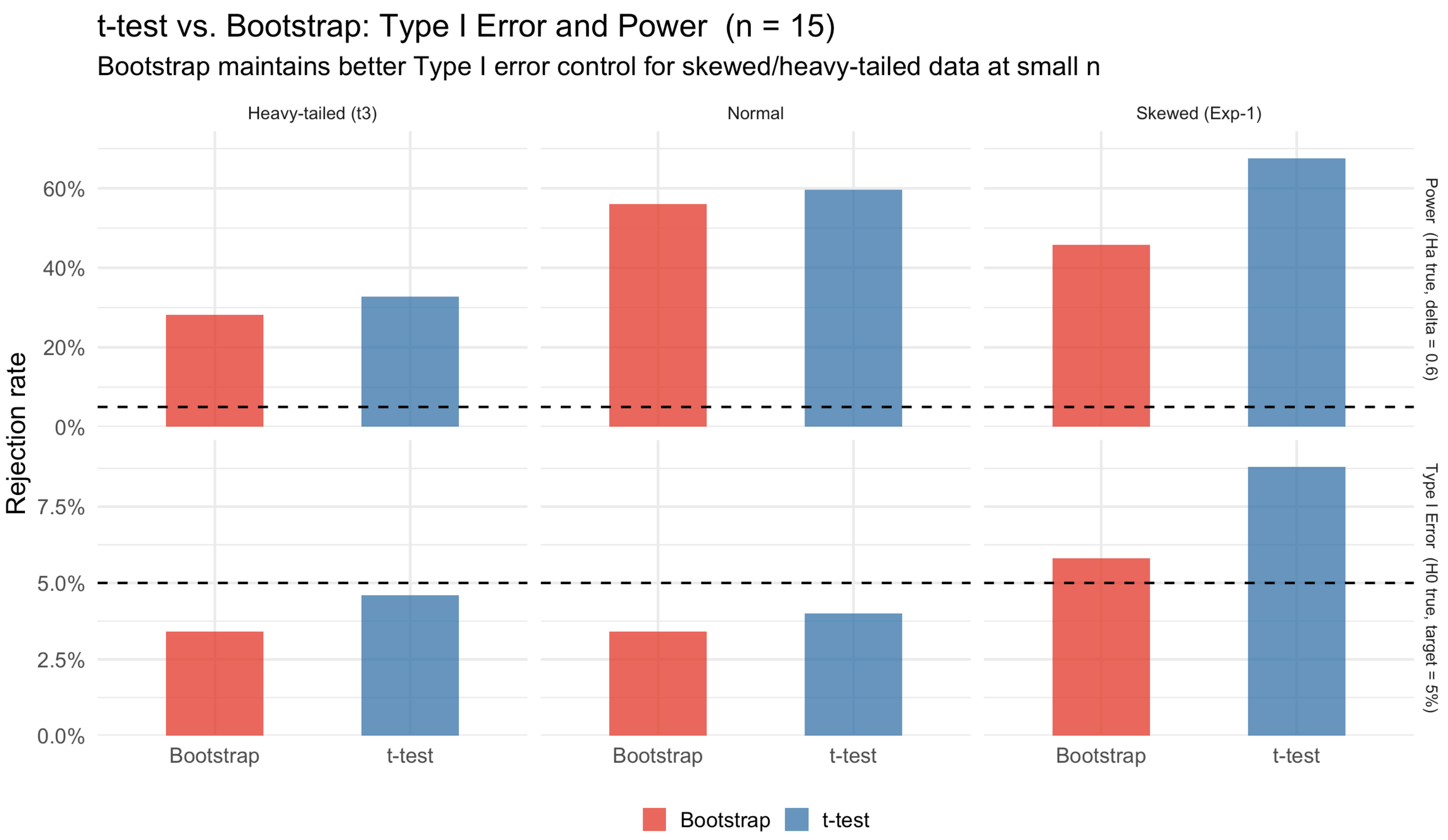
Key Message

The **bootstrap test** is the most flexible choice for one-sample problems with non-normal, skewed data — it only requires iid sampling and makes no symmetry assumption. The **t-test** is optimal when normality holds. Permutation methods will be covered after two-sample inference.

Simulation Study: Comparing t-test and Bootstrap

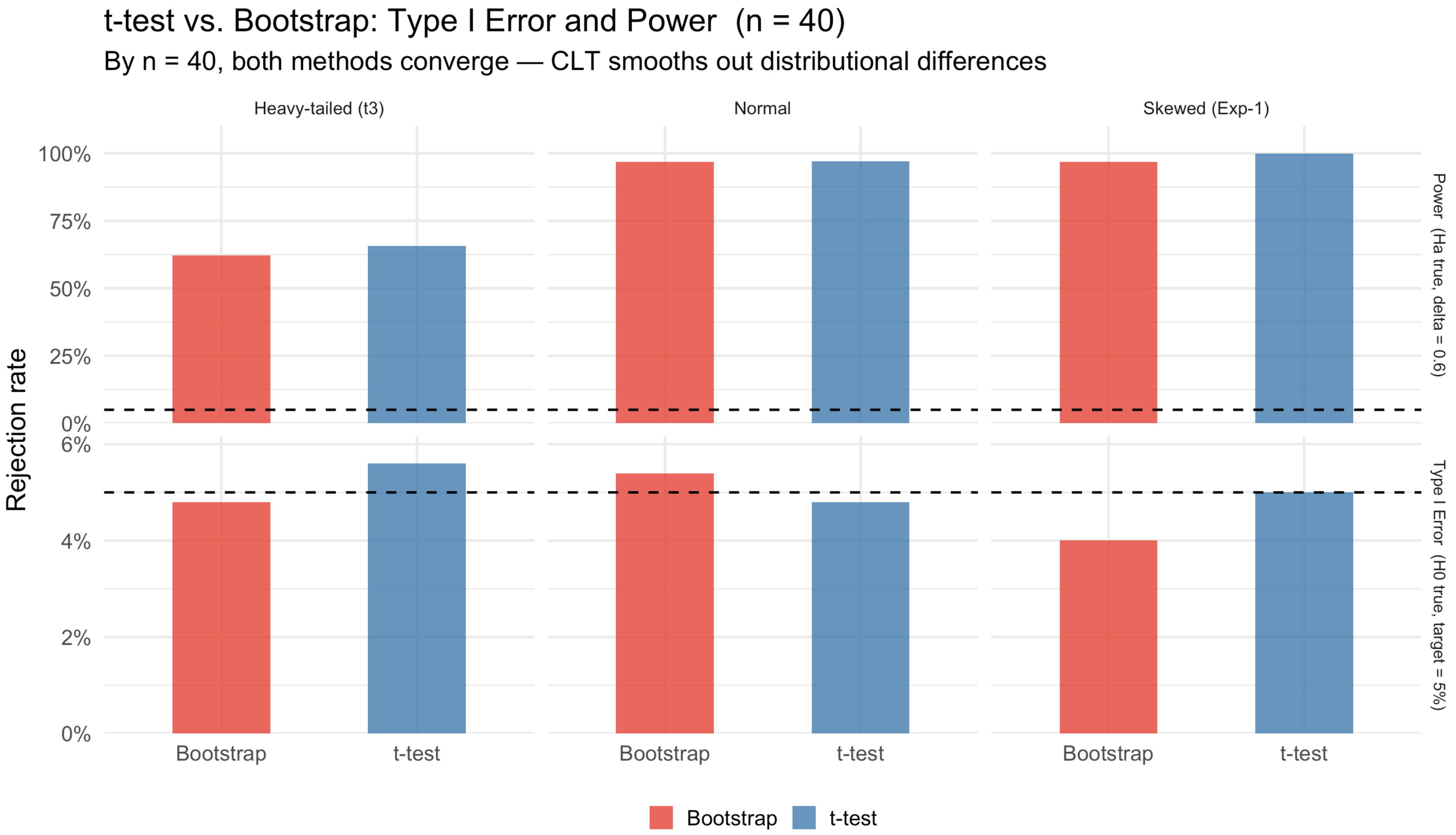
How well does each method control Type I error, and what is the power, across different population shapes?

▶ Code



Simulation: Larger Sample (n = 40)

▶ Code



Key Takeaways from Simulation

- **Normal data:** Both methods control Type I error well at $n = 15$ and $n = 40$; t-test has slightly higher power (it's optimal here)

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Practical Implication

For small samples ($n < 30$) with non-normal data, the bootstrap test is preferable. For $n \geq 40$, the t-test is nearly equivalent in practice.

Bootstrap Tests and the Duality with CIs

Bootstrap Tests and Bootstrap CIs: Are They Dual?

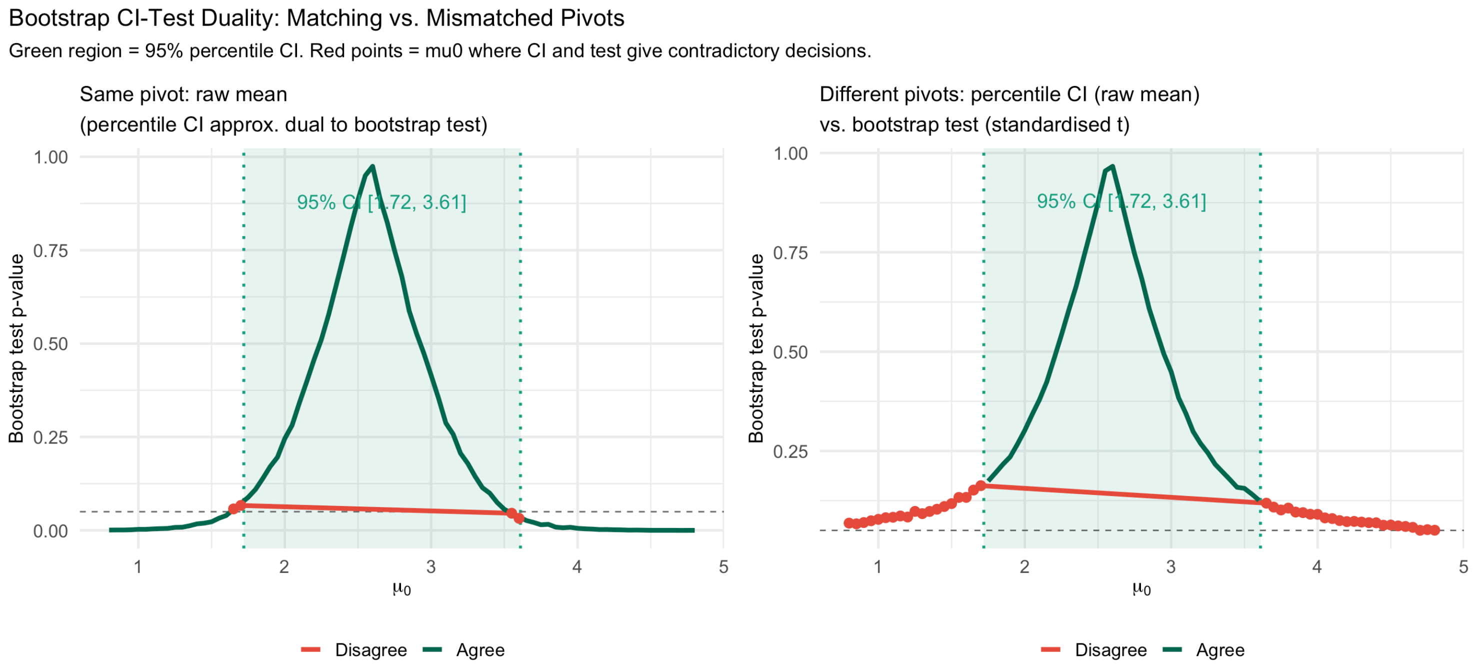
In the parametric setting, confidence intervals and tests are **exactly dual**. For bootstrap methods this is only **approximate** — and only when the CI and test use the **same pivot**.

The Key Condition

- **Percentile CI** resamples from the original data and uses $\bar{X}^*$ as the pivot
- **Bootstrap test** resamples from the shifted data and uses some test statistic T^*
- If both use the **same pivot** (e.g., raw mean), duality is approximate with only boundary disagreements from Monte Carlo noise
- If they use **different pivots** (e.g., percentile CI with raw mean vs. test with standardised t), they can disagree systematically over a wide range

Demonstrating: Same Pivot vs. Different Pivots

▶ Code



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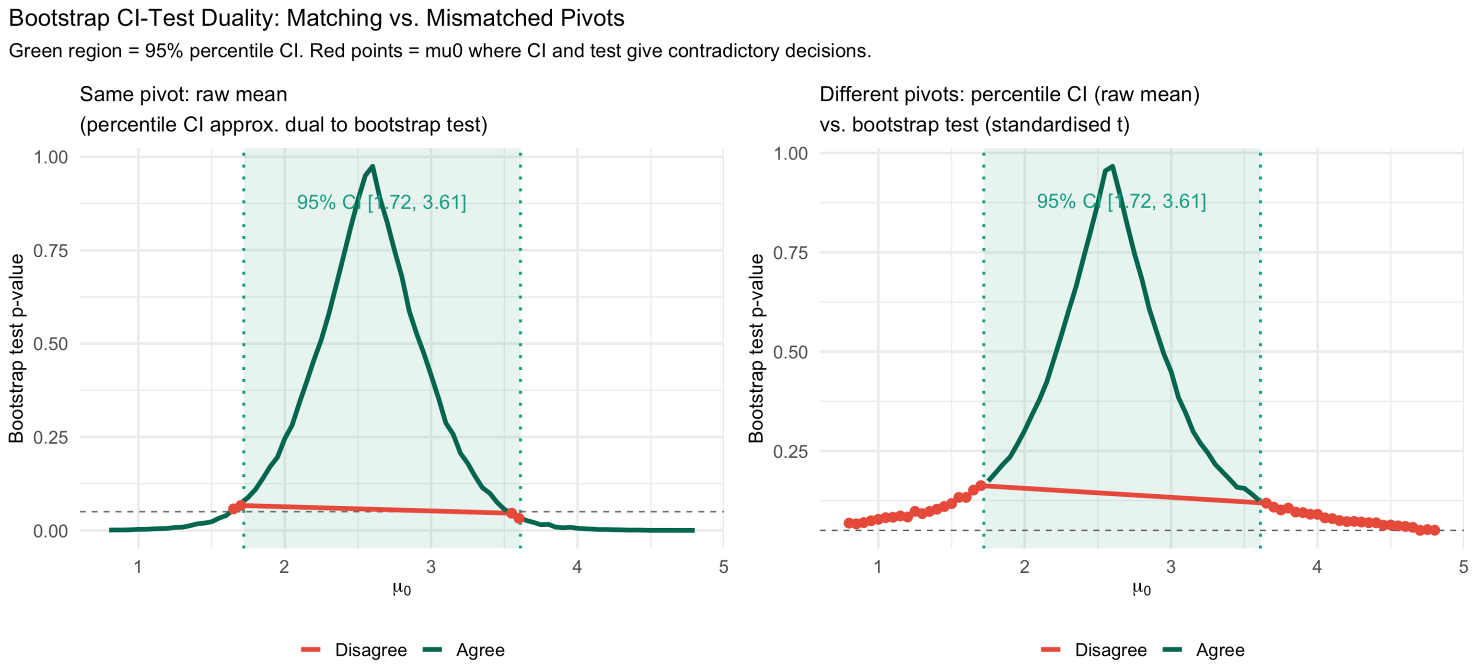

> ⚠ **Key Lesson**
>
> **Left panel:** Using the same pivot (raw mean), disagreements are rare and only occur right at the CI boundary due to Monte Carlo variability — duality is approximately satisfied.
> **Right panel:** Using different pivots (percentile CI vs. standardised t-test), disagreements are systematic and spread over a range — this is a methodological mismatch, not boundary noise.

Key Takeaways

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Today's Main Points (C&H Chapter 8)

1. **Bootstrap hypothesis tests** approximate the null distribution by resampling from data **shifted to satisfy** H_0 — this is fundamentally different from bootstrap CIs (which resample from the original data)
2. **The naive bootstrap is wrong:** resampling from the original data without shifting produces a distribution centered at $\bar{x}$, not at μ_0 — it does not represent the null distribution
3. **One-sample bootstrap test:** shift data so $\bar{x}^* = \theta_0$, resample, compute the t-statistic; the p-value is the fraction of bootstrap statistics as extreme as observed
4. **Bootstrap vs. t-test:** the bootstrap requires only iid sampling and no distributional assumptions; benefits are largest at small n with skewed or heavy-tailed data; both converge for large n
5. **Duality is approximate, not exact:** the bootstrap percentile CI and bootstrap t-test use different pivots, so near CI boundaries with small, skewed samples they can give contradictory decisions — reject by one criterion but not the other

Looking Ahead

Next class: Two-Sample z and t Tests; Confidence Intervals

- Two-sample z-test for means (σ known)
- Welch t-test for means (σ unknown)
- Pooled t-test (equal variance assumption)
- Two-sample inference for proportions

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Later: Permutation and Bootstrap Tests for Two Samples (C&H Ch. 8)

- Label-permutation tests for two independent groups
- One-sample sign-flip permutation test
- Bootstrap two-sample test

References

- Chihara, L. M. and Hesterberg, T. C. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 8: Bootstrap Hypothesis Testing.
- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 9.6.
- Efron, B. and Hastie, T. (2016). *Computer Age Statistical Inference*. Cambridge University Press. Chapter 15.

Questions?

Thank you!